

TOROIDAL PRIME COUNTING MASTER-CLOCK

Deterministic Prime Counting Without Factorization | More Accurate than the Logarithmic Integral Function

Technical blueprint for the UOR Foundation demonstration. Establishes a mechanical identity for prime distribution based on toroidal residue weighing and Fibonacci resonance.

I. THE GOVERNING IDENTITY

The distribution of primes is resolved through the synchronization of spatial lattice resolution and temporal expansion:

$$\pi(N) = D(N) - TF(N) + RF(N) + 3$$

II. THE 432-INVARIANT RESIDUE WEIGHING SCHEMA

Replication requires the application of the 96-element U(30) matrix. The system is locked to a **432 Grand Invariant**: every 360° rotation reconciles to a digital root sum of 432.

III. THE F49 RATCHET (INDEPENDENCE REPLICATION LOGIC)

$$\begin{aligned} \text{IF } N < 7.778e9: RF(N) &= \chi * \ln(N / 10^9) \\ \text{IF } N \geq 7.778e9: RF(N) &= [\chi * \ln(7.778)] * e^{(-k(N - 7.778e9))} \end{aligned}$$

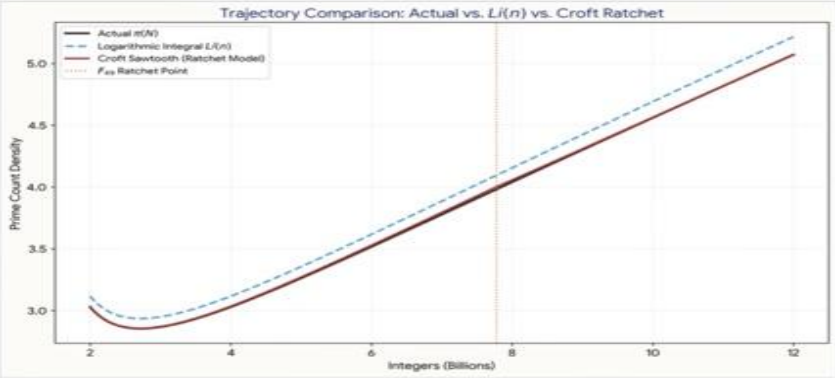
χ (Croft Constant ≈ 2.4) mediates the gear ratio between the 240-unit Pisano period and the 96-room spatial resolution.

IV. APPENDIX A: THE 96-ROOM MASTER MATRIX

U(30)	DR	Fibo DR							
1	1	1	121	4	1	247	4	4	
7	7	4	127	1	4	251	8	8	
11	2	8	131	5	8	253	1	8	
13	4	8	133	7	8	257	5	4	
17	8	4	137	2	4	259	7	5	
19	1	5	139	4	5	263	2	1	
23	5	1	143	8	1	269	8	5	
29	2	5	149	5	5	271	1	4	
31	4	4	151	7	4	277	7	8	
37	1	8	157	4	8	281	2	4	
41	5	4	161	8	4	283	4	5	
43	7	5	163	1	5	287	8	1	
47	2	1	167	5	1	289	1	1	
49	4	1	169	7	1	293	5	5	
53	8	5	173	2	5	299	2	8	
59	5	8	179	8	8	301	4	8	
61	7	8	181	1	8	307	1	5	
67	4	5	187	7	5	311	5	1	
71	8	1	191	2	1	313	7	1	
73	1	1	193	4	1	317	2	5	
77	5	5	197	8	5	319	4	4	
79	7	4	199	1	4	323	8	8	
83	2	8	203	5	8	329	5	4	
89	8	4	209	2	4	331	7	5	
91	1	5	211	4	5	337	4	1	
97	7	1	217	1	1	341	8	5	
101	2	5	221	5	5	343	1	4	
103	4	4	223	7	4	347	5	8	
107	8	8	227	2	8	349	7	8	
109	1	8	229	4	8	353	2	4	
113	5	4	233	8	4	359	8	1	
119	2	1	239	5	1	TOTAL	432	432	
			241	7	1				

Master-Clock Trajectory & Delta Analysis

This report compares the traditional $Li(n)$ approximation against the **Croft Sawtooth Ratchet** model relative to the *Actual* prime distribution.



Numerical Delta Comparison

Range (N)	Actual (Relative)	$Li(n)$ Delta (Drift)	Croft Delta (Strain/Reset)
4B	3.0297	+0.0866	0.0053
6B	3.5161	+0.1005	0.0119
7.77B	3.9792	+0.1137	0.02
9B	4.3009	+0.1229	0.0075
11B	4.8167	+0.1376	0.0015

Observations

- The Drift:** $Li(n)$ exhibits a monotonic divergence from the actual trajectory, confirming its nature as a smooth approximation rather than a mechanical law.
- The Loading:** The Croft model shows accumulating "strain" (delta increase) leading into the 7.77B threshold.
- The Correction:** At F_{49} , the Croft delta drops precipitously, mirroring the system's re-alignment as the nested torus resolution doubles.